
Toward a Temporal-Semantic Foundation Model for Event Prediction

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Real-world organizations generate large-scale streams of timestamped events,
2 where each event contains both semantic information, such as a textual description,
3 and temporal information, such as the time at which the event occurred. In collabora-
4 tion with KBC, this project investigates how to build a foundation model for such
5 event streams: a model that can learn general representations of historical events
6 and support downstream tasks that require predicting both what event will happen
7 next and when it will happen.

8 This setting combines two traditionally separate modeling problems. On the one
9 hand, language models provide strong tools for representing event descriptions, but
10 they usually treat sequences as ordered text and do not explicitly model irregular
11 time gaps between events. On the other hand, Temporal Point Processes model
12 the timing of future events, but often assume a fixed and limited set of event types,
13 making them less suitable for rich textual event descriptions. A central challenge is
14 therefore how to tokenize both events and time in a way that is scalable, sufficiently
15 granular, and informative for KBC-relevant prediction tasks.

16 The goal of this project is not only to train a predictive model, but also to systemat-
17 ically assess whether existing approaches for event modeling, time representation,
18 and Temporal Point Processes are appropriate for this use case. We study how
19 different event-tokenization and time-tokenization strategies affect representation
20 quality, temporal prediction, and downstream utility. By doing so, the project aims
21 to define the modeling requirements for a practical temporal-semantic foundation
22 model and identify the limitations of current methods when applied to real-world
23 KBC event streams.

24 1 Introduction

25 Modern organizations generate large-scale streams of timestamped events. Each event usually con-
26 tains both semantic information, such as a textual description or metadata, and temporal information,
27 such as the time at which it occurred. In collaboration with KBC, this project studies how to build a
28 foundation model for such event streams: a model that can predict both *what* is likely to happen next
29 and *when* it is likely to happen.

30 This problem combines two different modeling challenges. Language models are effective for
31 representing textual event descriptions, but they usually do not explicitly model irregular time gaps
32 between events. Temporal Point Processes (TPPs), on the other hand, provide a principled framework
33 for modeling event times, but often assume a fixed and limited set of event types. This makes them
34 less suitable for real-world settings where events may be described by rich text, high-cardinality
35 attributes, and heterogeneous metadata.

A central question in this project is therefore how to represent events and time in a scalable and informative way. Event tokenization determines how raw event content is converted into model inputs, while time tokenization determines how timestamps, time gaps, periodicity, and temporal intensity are exposed to the model. These choices directly affect the model’s granularity, scalability, and usefulness for KBC-relevant downstream tasks.

Formally, given a history of timestamped events

$$\mathcal{H}_i = \{(x_1, t_1), (x_2, t_2), \dots, (x_i, t_i)\},$$

where x_j is the event content and t_j is its timestamp, the goal is to learn a model that predicts both the next event x_{i+1} and its occurrence time t_{i+1} .

The goal of this project is to assess whether existing sequence models, time-aware language models, and TPP-based methods are suitable for this use case. By analyzing their assumptions and limitations, we aim to define the modeling requirements for a practical temporal-semantic foundation model over KBC event streams.

2 Background: Temporal Point Processes

Temporal Point Processes (TPPs) are probabilistic models for event sequences that occur in continuous time. Unlike standard discrete sequence models, which mainly model the order of events, TPPs explicitly model the time at which each event occurs [7]. A marked TPP represents a sequence as

$$\mathcal{H}_n = \{(t_1, e_1), (t_2, e_2), \dots, (t_n, e_n)\},$$

where t_i is the timestamp of the i -th event and e_i is its mark, usually interpreted as the event type. The goal is to model the distribution of the next event and its time given the history, i.e.,

$$p(t_{n+1}, e_{n+1} \mid \mathcal{H}_n).$$

A common way to define a TPP is through the conditional intensity function

$$\lambda^*(t, e) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(\text{event of type } e \text{ occurs in } [t, t + \Delta t) \mid \mathcal{H}_t^-)}{\Delta t},$$

where $\mathcal{H}_t^-$ denotes the event history strictly before time t . Intuitively, the intensity measures the instantaneous rate at which an event of type e is expected to occur, conditioned on the past. A high value of $\lambda^*(t, e)$ means that, according to the model, an event of type e is likely to occur soon after time t .

Given an observed event sequence over an observation window $[0, T]$, the standard TPP log-likelihood is

$$\mathcal{L} = \sum_{i=1}^n \log \lambda^*(t_i, e_i) - \int_0^T \sum_{e \in \mathcal{E}} \lambda^*(s, e) ds.$$

The first term rewards the model for assigning high intensity to observed events at their true times. The second term penalizes the model for assigning too much total intensity over the full observation window. This balance is important: the model should predict real events, but should not predict that events are likely everywhere all the time.

Neural TPPs replace hand-designed intensity functions with neural networks. Typically, an encoder such as an RNN, Transformer, or attention-based model first summarizes the history into a hidden representation h_n . This representation is then used to predict either the conditional intensity $\lambda^*(t, e \mid h_n)$ or the conditional density $p(t_{n+1}, e_{n+1} \mid h_n)$. This makes neural TPPs flexible enough to model complex temporal patterns such as burstiness, periodicity, and long-range dependencies.

For our KBC setting, the main limitation of classical marked TPPs is that the mark e_i is usually assumed to come from a fixed and relatively small set of event types. In contrast, real KBC events may contain rich textual descriptions, metadata, and high-cardinality attributes. Therefore, the event should be viewed not only as a discrete mark e_i , but as a richer semantic object x_i . The modeling target becomes

$$p(t_{n+1}, x_{n+1} \mid \mathcal{H}_n),$$

75 where x_{n+1} may include textual event content, categorical attributes, and other business-relevant
76 fields.

77 Recent work on language-model-based TPPs suggests one possible direction for this problem.
78 Instead of treating time and event type as separate specialized inputs, continuous time intervals
79 can be converted into tokens and combined with textual event information inside a language-model
80 framework. This is especially relevant for our project because it connects the TPP objective of
81 predicting *when* an event happens with the language-model objective of predicting *what* event happens.
82 However, it also highlights the central design question of this project: how should event content and
83 time be tokenized so that the representation is scalable, granular, and useful for downstream KBC
84 tasks?

85 3 Related Work

86 **Neural temporal point processes.** Temporal Point Processes (TPPs) provide the standard proba-
87 bilistic framework for modeling event sequences that occur in continuous time. The review of neural
88 TPPs by Shchur et al. [7] is relevant to our project because it defines the main modeling components
89 used in this literature: representing each event as a feature vector, encoding the event history, and
90 predicting the conditional distribution of the next event time and mark. This formulation fits our goal
91 of predicting both what event will happen next and when it will occur. However, most classical and
92 neural TPP formulations assume that each event is represented by a fixed mark, usually a categorical
93 event type. This is limiting for KBC event streams, where events may contain textual descriptions,
94 heterogeneous metadata, numerical fields, and high-cardinality attributes. The review also highlights
95 that TPP evaluation is not yet fully standardized and that metrics such as negative log-likelihood may
96 hide the separate quality of time and mark prediction. This creates a gap for our setting: we need a
97 framework that keeps the probabilistic temporal modeling strength of TPPs while supporting rich
98 event semantics and evaluation criteria that are meaningful for KBC downstream tasks.

99 **TPP-LLM.** TPP-LLM [4] is one of the first works to explicitly combine large language models
100 with temporal point processes. Instead of representing event types only as categorical identifiers,
101 the model uses textual descriptions of event types and combines them with temporal embeddings.
102 This direction is highly relevant for our project because it shows that pretrained language models can
103 improve event-sequence modeling when the event labels carry semantic meaning. It also demonstrates
104 the usefulness of parameter-efficient fine-tuning, such as LoRA, for adapting LLMs to temporal
105 event prediction. However, the method still mainly assumes that each event is described by an event
106 type text, rather than a rich multi-field business record. The next event type is predicted with a
107 classification head over a fixed event vocabulary, and the temporal component is handled through
108 additional embeddings and TPP-style losses. Therefore, the gap for our use case is that KBC events
109 may require more general event tokenization, including free text, categorical fields, numerical values,
110 and possibly structured key-value attributes, rather than only textualized event types.

111 **Language-TPP.** Language-TPP, introduced in Byte-token Enhanced Language Models for Tem-
112 poral Point Processes Analysis [2], moves further toward a unified language-model formulation of
113 TPPs. Its key idea is to convert continuous time intervals into byte-level tokens and insert them into a
114 textual event template together with event type and description tokens. This is directly relevant to
115 our project because it treats time tokenization as a central design problem, which is also one of our
116 main questions. By representing each floating point time interval using byte tokens, the model can
117 perform time prediction, type prediction, and description generation using a standard autoregressive
118 language-model interface. The weakness is that this design can cause long-context problems when
119 event descriptions are long or when histories contain many events. The authors also note potential
120 scalability issues for very large event sequences and richer modalities. For KBC, this suggests that
121 byte-token time encoding is a strong baseline to evaluate, but it may not be sufficient alone: we
122 still need to study whether byte-level time tokens preserve the right granularity, whether they are
123 robust for business time scales, and whether the resulting sequence length is practical for large-scale
124 deployment.

125 **Long-range modeling of multimodal event sequences.** The work on Long-range Modeling and
126 Processing of Multimodal Event Sequences [3] extends Language-TPP to multimodal event streams.
127 It proposes MM-TPP, a framework that jointly models temporal, categorical, textual, and visual event

information, and introduces an adaptive compression mechanism based on temporal similarity. This paper is relevant because it addresses a key practical issue for our project: real event histories can become too long for standard Transformer attention, especially when each event contains rich content. The compression mechanism replaces temporally similar consecutive events with a special token, allowing the model to use a longer effective history within a fixed context window. However, the compression criterion is mainly based on similarity between adjacent inter-event intervals. For KBC, this may be too narrow: two events may have similar timing but very different business meaning, or different timing but similar semantic role. Therefore, the gap is to design compression or aggregation methods that consider both temporal similarity and semantic/business similarity, rather than relying only on local time-interval patterns.

PRAGMA. PRAGMA [5] is the closest work to our intended application domain because it builds a foundation model for multi-source banking event sequences. The model uses a key-value-time tokenization scheme with type-specific value encodings for numerical, categorical, and textual fields, and is pretrained with masked modeling over heterogeneous user histories. This is highly relevant for KBC because it shows that banking event streams can be modeled directly as structured event histories rather than flattened into raw text. It also motivates the importance of preserving field identity, numerical meaning, timestamps, static profile state, and heterogeneous event sources. However, PRAGMA is primarily an encoder-style representation model for downstream discriminative tasks such as credit scoring, fraud detection, product recommendation, and lifetime value prediction. It is not mainly designed as a generative model that predicts the next event description and its occurrence time. In addition, PRAGMA processes event histories mostly at the individual record level, and the paper reports limitations for highly relational tasks such as anti-money laundering. This leaves an important gap for our project: we need to assess whether a PRAGMA-style tokenization scheme can be combined with a TPP-style or autoregressive objective to support generative next-event and next-time prediction in KBC event streams.

Positioning of our project. Together, these works suggest that no existing approach fully matches the KBC use case. Neural TPPs provide the mathematical foundation for continuous-time event prediction, but usually assume relatively simple event marks. TPP-LLM and Language-TPP show how language models can incorporate event semantics and time tokens, but they do not fully solve the problem of scalable, heterogeneous business-event tokenization. MM-TPP addresses long-context and multimodal event sequences, but its compression strategy is mainly temporal and may not preserve business semantics. PRAGMA provides the most relevant foundation-model design for banking event histories, but it is primarily an encoder model for discriminative tasks rather than a generative temporal-semantic model. Our project is positioned at the intersection of these directions: we aim to study how to tokenize heterogeneous KBC events and time in a scalable and granular way, and to evaluate whether current TPP, LLM-based TPP, and banking foundation-model approaches are sufficient for predicting both future event content and future event timing.

3.1 Architecture Alternatives to Full LLM Training

3.1.1 The NVIDIA Transaction Foundation Model Blueprint github

A highly modular custom tokenizer powered, built on top of common frameworks, e.g. Pytorch, XGB, SKLearn, etc.. This tokenizer converts heterogeneous tabular fields such as merchant categories, transaction amounts, and time deltas into domain-specific token sequences, efficiently handling the MMTT structure problem natively.

They provide a pre-trained checkpoint: a decoder-only foundation model, trained via causal language modeling.

3.1.2 IBM TabFormer for Tabular Time Series

[6]

IBM’s TabFormer proposes highly effective neural network architectures designed specifically for tabular time series data. TabFormer offers two distinct architectural paradigms: one designed for learning representations that is analogous to BERT (which can be pre-trained end-to-end and utilized

178 in downstream tasks like credit fault prediction), and one that is akin to GPT (utilized for the
179 generation of realistic synthetic tabular sequences).

180 3.1.3 Contrastive Learning for Event Sequences (CoLES)

181 [1]

182 Instead of utilizing compute-heavy causal autoregressive generation (which requires predicting every
183 subsequent token) or Masked Language Modeling (which requires massive bidirectional attention
184 matrices), CoLES utilizes a highly efficient contrastive loss methodology.

185 The mechanism works by taking a single user’s transaction sequence and splitting it into sub-
186 sequences to generate positive pairs. It then contrasts these positive pairs against negative pairs
187 generated from entirely different users. By forcing the encoder to pull sub-sequences of the same
188 user closer together in the embedding space while pushing sub-sequences of different users apart, the
189 model learns incredibly robust, user-level embeddings with a fraction of the compute required for
190 standard token prediction.

191 **PyTorch-Lifestream github** Developed specifically for processing terabyte-scale volumes of
192 discrete event sequences (including credit card transactions, purchase histories, and clickstreams),
193 ptls serves as an accessible, scalable alternative to building complex NeMo or Megatron pipelines.

194 3.1.4 Mambular (State Space Models Mamba) github

195 [8]

196 The Mamba architecture provides a drop-in replacement for attention mechanisms. Mamba blocks
197 process data sequentially with linear scaling, capturing relationships feature-by-feature analogous
198 to RNNs, but with advanced, hardware-aware parallelization that prevents the vanishing gradient
199 problems associated with traditional recurrent architectures. The Mambular library, part of the broader
200 DeepTab open-source Python package, brings the power of the Mamba architecture directly to tabular
201 deep learning, offering a suite of models for regression and classification tasks.

202 3.1.5 Open Benchmarks and Synthetic Generators

203 The Multimodal Banking Dataset (MBD) github

204 The Multimodal Banking Dataset (MBD) provides an unprecedented, industrial-scale open-source
205 benchmark constructed from real, proprietary, yet fully anonymized bank data.

206 Multi-Agent Synthetic Financial Data Generators

207 PaySim (mobile money transactions), BankSim (bank payments based on aggregated Spanish banking
208 data), IBM AML Data (Anti-Money Laundering (AML) data generated using a detailed multi-agent
209 virtual world model).

210 4 Questions

- 211 1. What are the relevant tasks you want to perform with the model? For example: prediction,
212 embeddings, generation, or another downstream task.
- 213 2. What is the required context length? How many events should the model observe, and how
214 long is the textual description of each event?
- 215 3. Is the event space closed or open? That is, do we have a fixed set of N possible event types,
216 or are events described using free-text descriptions?
- 217 4. How much data is available, and what computational resources do we have? For example:
218 dataset size, number of GPUs, GPU memory, and expected training budget.
- 219 5. What is the format of the data? For example: JSON, dictionaries, tabular format, or another
220 structured representation.

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